

KG-o1: Enhancing Multi-hop Question Answering in Large Language Models via Knowledge Graph Integration

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Abstract

Large Language Models (LLMs) face challenges in knowledge-intensive reasoning tasks like classic multi-hop question answering, which involves reasoning across multiple facts. This difficulty arises because the chain-of-thoughts (CoTs) generated by LLMs in such tasks often deviate from real or a priori reasoning paths. In contrast, knowledge graphs (KGs) explicitly represent the logical connections between facts through entities and relationships. This reflects a significant gap. Meanwhile, large reasoning models (LRMs), such as o1, have demonstrated that long-step reasoning significantly enhances the performance of LLMs. Building on these insights, we propose **KG-o1**, a four-stage approach that integrates KGs to enhance the multi-hop reasoning abilities of LLMs. We first filter out initial entities and generate complex subgraphs. Secondly, we construct logical paths for subgraphs and then use knowledge graphs to build a dataset with a complex and extended brainstorming process, which trains LLMs to imitate long-term reasoning. Finally, we employ rejection sampling to generate a self-improving corpus for direct preference optimization (DPO), further refining the LLMs' reasoning abilities. We conducted experiments on two simple and two complex datasets. The results show that KG-o1 models exhibit superior performance across all tasks compared to existing LRMs. The datasets, models, and code for KG-o1 are publicly available¹.

Introduction

Multi-hop question answering involves synthesizing information from multiple facts through a sequence of logical steps to derive the correct answer (Trivedi et al. 2020; Khashabi et al. 2020). For instance, answering the question “*Who is the spouse of the actor who played Batman in The Dark Knight*” requires first identifying the actor who played Batman in The Dark Knight, and then determining the spouse of that actor. Although large language models have demonstrated impressive performance in question-answering tasks (Kamalloo et al. 2023; Wang and Qin 2024), they still struggle to perform correct and logical reasoning in complex scenarios

(Kaplan et al. 2020; Wei et al. 2022; Zheng et al. 2023; Ding et al. 2024).

Meanwhile, Knowledge Graphs (KGs) holding the potential for handling multi-hop questions, which inherently encompass logical reasoning paths between facts. For example, to answer the question above, one can follow a reasoning path between the two facts: “(*The Dark Knight*, *hasActor*, *Christian Bale*) $\rightarrow$ (*Christian Bale*, *spouse*, *Sibi Blazic*)”, to arrive at the correct answer. Thus, in this paper, we focus on leveraging the structured nature of KGs to guide LLMs along explicit reasoning paths, thereby enhancing their performance on complex multi-hop question answering tasks (Qu et al. 2020; Choudhary and Reddy 2023).

Although previous studies have explored leveraging KGs to enhance the multi-hop reasoning capabilities of LLMs, they have not fundamentally improved the models' intrinsic ability to reason over multiple facts and generate reasoning processes aligned with a priori reasoning paths. Instead, KGs are typically treated as external resources. These approaches either incorporating structured data to guide chain-of-thought generation (Chen et al. 2020; ZHAO et al. 2024) or adopting retrieval-augmented generation to answer complex questions (Trivedi et al. 2023; Park et al. 2023; Tang and Yang 2024). Both paradigms rely on separate retrieval modules or symbolic systems, offering only limited improvements to the underlying reasoning abilities of LLMs. Meanwhile, OpenAI's breakthrough with o1² has shown that long stepwise thinking significantly improves the reasoning capabilities of LLMs (Guan et al. 2024; Zhong et al. 2024a). LRMs employ a test-time compute scaling paradigm (Snell et al. 2024; Qi et al. 2024; Xi et al. 2024) to simulate slower, more deliberate Human System 2 thinking (Guan et al. 2025), which improve the accuracy and flexibility of reasoning. Various methods have been proposed to in scientific, mathematical, programming, and medical reasoning tasks (Zhao et al. 2024; Min et al. 2024; Chen et al. 2024; Guan et al. 2025; Xiang et al. 2025). Despite these advances, the potential for knowledge-intensive multi-hop reasoning tasks is still underexplored.

To this end, we propose **KG-o1**, which integrates KGs to

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¹<https://github.com/JOHNNY-fans/KG-o1#>

²<https://openai.com/o1/>

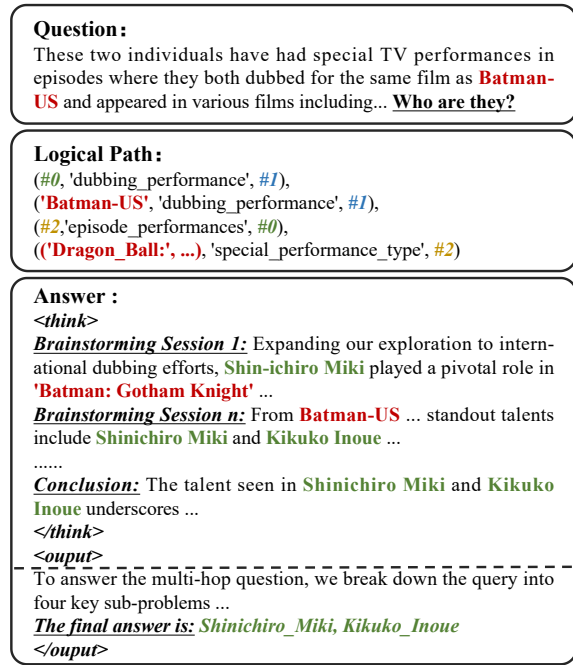


Figure 1: Example of KG-MHQA. The answer has two parts: *Think*, which explicitly integrates the logical path, and *Output*, which uses this reasoning to solve the multi-hop question.

evolve an LLM into an LRM, thereby enhancing multi-hop question answering in complex scenarios. Our approach involves extracting explicit reasoning paths from knowledge graphs and incorporating their logic into the reasoning chains of LLMs through training. As illustrated in Figure 1, we internalize the logical paths from KGs into the model’s intrinsic reasoning chains. The explicit process is as follows:

First, we consider relationship types to select complex subgraphs from the knowledge graph. Second, we convert these subgraphs into clear logical reasoning paths and identify entities that can be questioned. Next, the selected entities and reasoning paths are provided to ChatGPT-4o (OpenAI 2024) to generate multi-hop questions, resulting in complex multi-hop QA pairs. Based on these pairs, we employ data distillation to iteratively generate extended reasoning processes, yielding the KG-MHQA SFT dataset. Through supervised fine-tuning (SFT), base models are able to emulate a slow thinking paradigm, resulting in the KG-o1 SFT models. Finally, we propose the “Self-improved Adaptive DPO” strategy, utilizing the KG-o1 SFT models and an advanced LRM to obtain positive and negative responses through rejection sampling, thereby constructing the adaptive KG-MHQA DPO dataset. Direct preference optimization (DPO) is then applied to align the SFT models with more appropriate reasoning processes, producing the final KG-o1 models. Our main contributions are summarized as follows:

- We propose KG-o1, a framework grounded in the idea of deeply integrating KGs with LLMs. This framework provides a reusable process that systematically evolves an LLM to an LRM based on reasoning paths of KGs,

thereby enhancing intrinsic multi-hop question answering for Large Language Models.

- We construct the KG-MHQA SFT dataset by leveraging logical paths from the FB15k knowledge graph, and propose the Self-improve Adaptive DPO strategy to build the KG-MHQA DPO dataset. Utilizing these datasets, we develop a series of KG-o1 models.
- Extensive experiments on several datasets and our self-built KG-MHQA test dataset demonstrate that our KG-o1 models achieve competitive performance compared to ChatGPT-4o and o1-mini. Furthermore, ablation and domain transferability experiments verify the strong generalization ability of our approach.

Related Work

Large Reasoning Models

Large reasoning models enhance the performance of LLMs by applying long-term reasoning at test time, in contrast to traditional pre-trained models that prioritize scalability through increased model size or expanded training data (Henighan et al. 2020; Yang et al. 2024; Zeng et al. 2024). Recent advancements have introduced models like OpenAI-o1 (Jaech et al. 2024), DeepSeek-R1 (Guo et al. 2025), and Gemini 2.0 Flash Thinking (Google DeepMind 2024), which have undergone large-scale SFT and reinforcement learning (RL). These models demonstrate that scaling at test time can significantly improve reasoning capabilities on complex tasks such as mathematics, coding, and scientific reasoning (Temsah et al. 2024; Zhang et al. 2024; Zhong et al. 2024b).

Recent research has advanced the reasoning capabilities of LLMs. Open-o1 (O1 2023) and DeepSeek-R1-Distill (Guo et al. 2025) leverage knowledge distillation with fine-grained filtering to provide high-quality process data for smaller LLMs. QWQ (Team 2024) and rStar Math (Guan et al. 2025) synthesize long reasoning via CoT, using self-evolved rewards and strategies like MCTS or recursive self-questioning to improve depth and accuracy. Recent work also refines long-form thinking quality: DRT-o1 (Wang et al. 2024) uses structured workflows for better iterative reasoning, while search-o1 (Li et al. 2025) incorporates retrieval-augmented methods to bridge knowledge gaps. For training, DPO (Rafailov et al. 2023) and PRM (Lightman et al. 2023) enhance inference via preference optimization and process supervision, while GRPO (Shao et al. 2024) and DAPO (Yu et al. 2025) further improve training efficiency and long-chain inference stability.

Multi-Hop Question Answering

Multi-hop question answering requires LLMs to reason across multiple facts. Notable datasets like HotpotQA, 2Wiki-MultihopQA, and MuSiQue (Yang et al. 2018; Ho et al. 2020; Trivedi et al. 2022) provide both relevant and distracting documents for robust evaluation. Recent extensions incorporate structured data, including KGs (Chen et al. 2020; Mihaylov et al. 2018; He et al. 2024) and scientific tables, while specialized domain corpora have been used in retrieval-augmented generation (RAG) systems (Tang and Yang 2024).

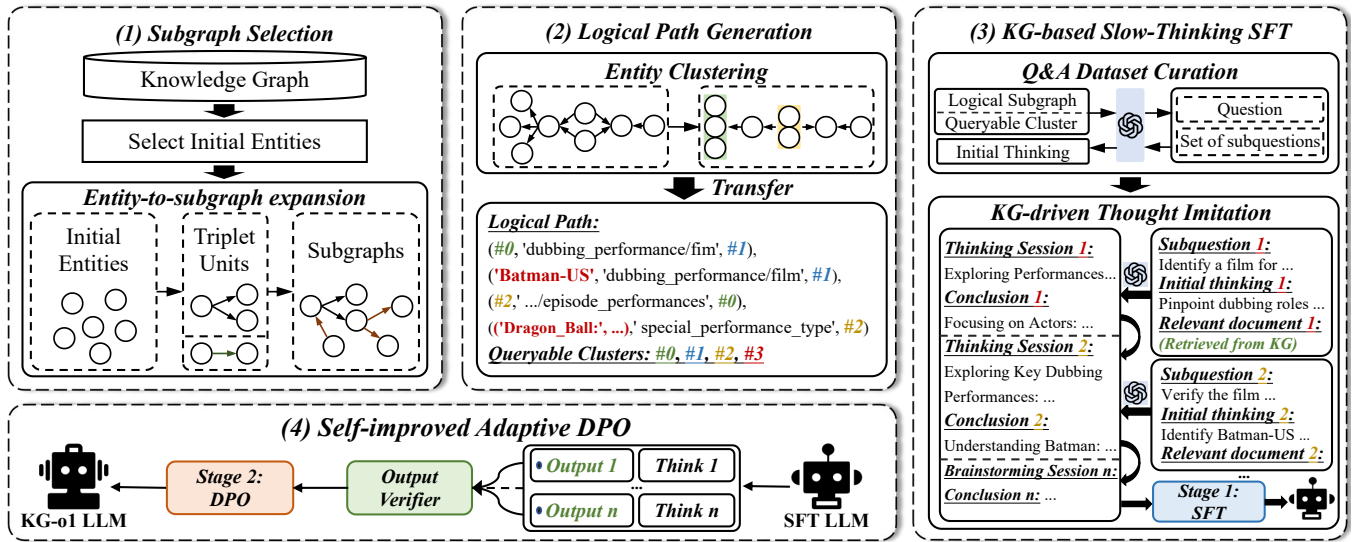


Figure 2: Overview of KG-o1 framework for enhancing multi-hop question answering in LLMs.

Recent advancements in LLM-based methods primarily treat KGs as external aids or use prompt engineering approaches, rather than enhancing the intrinsic reasoning abilities of LLMs. For example, CoT (Saparov and He 2023; Han et al. 2022) encourages step-by-step reasoning and has been adopted in methods such as PathFiD (Yavuz et al. 2022) and IRCot (Trivedi et al. 2023). Similarly, question decomposition approaches (Yavuz et al. 2022; Zhou et al. 2023) break down complex questions into simpler sub-questions, facilitating multi-step inference. RAG-based frameworks leverage iterative cycles of retrieval and reasoning to generate multi-hop answers by combining LLMs with retrieved evidence (Tang and Yang 2024; Park et al. 2023).

Methodology

As shown in Figure 2, our method consists of four main stages. (1) Subgraph Selection: We first filter out the initial entities and then use a one-to-many approach for entity-to-subgraph expansion. (2) Logical Path Generation: Next, we transform the subgraph into logical paths and select entities that can be questioned. (3) KG-based Long-term Thinking SFT: Then, we construct complex multi-hop QA pairs, iteratively generate the SFT dataset, and fine-tune LLMs to simulate slow thinking based on the logical paths. (4) Self-improved Adaptive DPO: Finally, we generate contrastive instances and apply direct preference optimization to align models with more appropriate reasoning processes.

SubGraph Selection

Select Initial Entities. In this module, we construct the initial entity set by applying two main criteria on the knowledge graph: (1) entities must be associated with multiple distinct types to ensure cross-domain coverage, and (2) entities must be connected to a diverse set of relations, which ensures the diversity and representativeness of the final subgraphs. In this paper, we use the widely adopted Freebase 15k knowledge

graph³ and randomly sampled 1,000 key entities as our initial entity set for data construction.

Entity-to-subgraph expansion. Given initial entities, two steps are performed to select factual subgraphs containing two and three relationship types from the knowledge graph, yielding 253,374 and 1,624,786 subgraphs, respectively.

First, each initial entity is expanded to generate triplet units, each unit consisting of one or more triplets with a single relation type. This process is formally expressed in Formula 1:

$$S_j = \bigcup_{|R|} \{s_r^{e_i}\}, e_i \in N, j = 1, \quad (1)$$

where N is the set of initial entities, e_i is the i -th entity, j denotes the number of relation types, and $s_r^{e_i}$ represents a triplet unit originates from entity e_i as its starting node.

Second, each triplet unit is iteratively expanded by adding triplets from the knowledge graph. The process is formally expressed in Formula 5:

$$S_j = S_{j-1} \cup \{(m, r, n_i) \text{ or } (n_i, r, m) | m \notin N, n_i \in N, r \in R\}, j > 1 \quad (2)$$

For each subgraph, all relation types R connected to its entities are identified. For each $r \in R$, triplets are added if they satisfy: (1) one entity is in the subgraph, (2) the other is in the global knowledge graph, and (3) the connecting relation is r . See Section “Entity-to-Subgraph Expansion Examples” in the Appendix for detailed examples.

Logical Path Generation

In this section, we aim to transfer the subgraphs into logical paths, which are then used to select the entities that can be questioned. An explicit sample is provided in Appendix Section “Logical Triplets Generation Examples”. Specifically, we cluster entities within each subgraph based on their connected relations: Two entities e_i and e_j are grouped into

³<https://huggingface.co/datasets/VLyb/FB15k>

the same cluster C_k if, for every relation $r \in R$, they are connected to exactly the same set of entities:

$$\forall r \in R, \text{Neigh}(e_i, r) = \text{Neigh}(e_j, r),$$

where $\text{Neigh}(e, r)$ denotes the set of entities linked to e by relation r in either direction (i.e., as subject or object). After clustering, we transform the subgraph into logical paths by replacing each entity cluster with identifiers (e.g., “#1”). Finally, we apply heuristic rules to exclude clusters that are not suitable as question targets: Specifically, a cluster is excluded if its entities can be uniquely identified using only a subset of the triplets in logical paths.

KG-based Long-term Thinking SFT

Existing advanced LRMs have demonstrated that LLMs can remarkably enhance their reasoning ability through long-term thinking. In this section, we first prompt ChatGPT-4o (OpenAI 2024) to generate multi-hop questions, resulting in complex multi-hop QA pairs. Next, we incorporate the a priori reasoning logic and knowledge from KGs to construct detailed long-term thinking processes for each QA pair. These reasoning chains are explicitly represented and concatenated to the final response using unique tokens, resulting in the KG-MHQA SFT dataset. Through the SFT process, the models effectively simulate and exhibit long-term thinking behaviors.

QA Dataset Curation. Based on the two categories of logical paths obtained in the Logical Path Generation Section, we aim to construct multi-hop QA pairs that require complex reasoning over high-density facts. To ensure the question diversity and mitigate overfitting, we employ a uniform sampling strategy, categorizing the sampled paths based on the number of their triplets and their in-degree and out-degree patterns. This process yields comprehensive logical paths containing 2 to 6 reasoning hops, which encompass 667 distinct structural types and cover 85% of entities. Afterward, we impose constraints regarding question targets, logical paths, and interrogative syntax to guide ChatGPT-4o (OpenAI 2024) in generating questions. To ensure question quality, ChatGPT-4o is further required to decompose each multi-hop question and verify that each sub-question strictly corresponds to the underlying logical path. Our prompts are exhibited in the Appendix Section “Dataset Curation Prompts”. The corresponding entities of the question targets are used as answers, resulting in complex multi-hop QA pairs.

KG-driven Thought Imitation. At this stage, we aim to construct KG-based long-term thinking SFT data for fine-tuning LLMs to imitate the long-term thinking paradigm. Specifically, we integrate KGs and employ an iterative approach to construct the long-term thinking process. Given the complex multi-hop QA pairs, we plan the iterative optimization sequence according to prior reasoning paths to ensure that the process adheres to the reasoning logic. To ensure that the reasoning process is guided by the provided knowledge graph, we retrieve relevant triples at each iteration, present them in textual form, and provide them to the large language model as references. This approach encourages the model to integrate internal knowledge with external information, facilitating a more thorough and comprehensive reasoning process.

Split	Total	2-Hop	3-Hop	4-Hop	5-Hop	6-Hop
Train	5030	576	771	938	1,416	1,329
Dev	456	103	110	115	113	15
Test	307	51	54	73	104	25

Table 1: Statistics of KG-MHQA SFT dataset.

The iterative procedure is formally defined in Formula 3:

$$P_n = f(Q_n, T_n, K_n, P_1, P_2, \dots, P_{n-1}), \quad (3)$$

where P_n is the refined thinking process at the n -th iteration, Q_n is the subquestion, T_n is corresponding initial thinking segment, and K_n is the knowledge for the current iteration. The previous iterations $P_1, P_2, \dots, P_{n-1}$ are used to refine the thinking in the current iteration.

After generating the long-term thinking process, we introduce special tokens to distinguish the thought process from the output. In the *think* part, the model engages in brainstorming and extensive reasoning, while in the *output* part, it summarizes the reasoning steps, derives a solution from the subquestions, and generates the final answer. Using these QA pairs, we construct the KG-MHQA SFT dataset and fine-tune LLMs, enabling them to effectively emulate long-term reasoning. The sample data and statistics of KG-MHQA SFT dataset are shown in Figure 1 and Table 1.

Self-improved Adaptive DPO

Direct Preference Optimization (DPO) has been demonstrated as an effective approach for reducing harmful outputs and improving generation quality in large language models. To improve the long-term thinking process quality of our KG-o1 SFT models, we implement DPO optimization after the supervised fine-tuning stage. Due to variations in training data and model architectures, the knowledge and reasoning capabilities of different models vary, making SFT data distilled from a single source LLM insufficient to achieve consistent improvements across models.

To address this limitation, we propose a “Self-improved Adaptive DPO” strategy that dynamically constructs contrastive positive and negative pairs by combining SFT data with KG-o1 SFT model responses. As shown in stage 2 of Figure 2, for each sample in the SFT dataset, we implement a dual path contrastive data construction protocol using the correctness of the answers as the gold standard: When the response generated by KG-o1 SFT model matches the correct answer, it is treated as a positive instance, with the corresponding SFT sample response as the negative. If the generated response is incorrect, the original SFT sample response remains positive, while the chain-of-thought response generated by ChatGPT-4o is used as the negative. This adaptive strategy ensures stable alignment of reasoning processes and response quality. The resulting KG-MHQA DPO dataset is then used for DPO optimization on the KG-o1 SFT model, aligning suitable long-term thinking processes for multi-hop question answering.

Model	Method	KG-MHQA †				2WikiMultiHopQA				HotpotQA				MINTQA			
		EM	F1	Prec.	Rec.	EM	F1	Prec.	Rec.	EM	F1	Prec.	Rec.	EM	F1	Prec.	Rec.
Large Reasoning Models																	
o1-mini	CoT	41.69	53.25	51.77	60.33	42.60	61.46	56.35	78.90	49.80	67.33	67.47	76.67	75.67	85.96	85.20	90.21
Open-o1	CoT	44.33	60.33	56.36	79.46	25.08	36.64	35.04	44.80	35.20	55.13	49.77	72.06	37.60	55.05	53.75	69.78
QwQ-32B	CoT	38.44	49.04	47.45	64.20	68.33	79.06	77.92	84.62	56.20	70.96	67.70	82.75	48.40	64.94	67.04	68.94
Gemini-2.0-FT	CoT	48.21	60.40	57.76	70.23	66.20	76.23	74.13	82.84	57.00	72.88	74.95	79.60	80.67	88.53	87.49	92.28
DeepSeek-R1	CoT	44.95	56.47	54.79	66.99	67.00	76.76	75.30	86.33	57.00	72.24	75.21	78.34	85.67	88.95	88.80	91.21
General Purpose Large Language Models																	
ChatGPT-4o-mini	CoT	33.22	42.94	42.04	48.55	62.40	72.93	71.50	78.02	53.20	69.50	71.73	71.47	78.00	83.24	82.96	84.96
ChatGPT-4o	CoT	42.02	55.53	53.10	65.88	70.00	81.02	78.73	86.95	56.40	72.56	74.66	77.81	78.33	87.42	86.21	90.94
Llama3.1-8B	CoT	23.45	36.50	34.78	56.97	36.59	46.84	45.40	56.59	42.00	55.54	56.98	65.94	54.00	71.26	68.84	82.68
	Ours	41.04	57.81	53.63	69.44	55.00	68.59	66.84	73.77	43.40	60.16	62.20	62.24	67.00	75.97	74.92	79.02
Qwen2.5-7B	CoT	14.98	24.34	22.86	48.57	29.00	44.50	40.17	70.07	13.35	25.90	23.39	75.51	21.67	33.33	29.58	85.50
	Ours	30.29	43.59	41.22	61.12	32.40	41.00	39.72	59.00	25.40	36.35	37.67	54.76	50.00	58.08	57.05	75.26
DeepSeek-14B	CoT	37.78	48.09	46.64	54.96	54.20	68.91	66.26	77.46	48.02	64.62	66.70	67.81	69.23	79.06	78.78	81.99
	Ours	34.31	45.81	43.78	57.73	45.89	62.86	59.23	74.98	44.17	62.25	62.10	72.49	64.66	75.67	74.60	84.99
Qwen2.5-14B	CoT	29.97	38.97	37.64	54.78	57.20	68.15	65.86	79.44	42.20	55.31	55.14	78.66	61.00	70.50	68.77	89.18
	Ours	45.60	69.36	63.75	84.46	62.40	74.45	72.56	79.45	51.00	67.11	69.66	68.18	72.33	79.37	78.73	81.49

Table 2: Exact-Match (EM), F1, Precision, and Recall results of Large Reasoning Models and General Purpose Large Language Models on multi-hop reasoning datasets. † represents our self-build dataset, Gemini-2.0-FT denotes Gemini-2.0-Flash-Thinking-exp, DeepSeek-14B denotes DeepSeek-R1-Distill-Qwen-14B.

Experiments

In this section, we evaluate the performance of prominent LRMs, advanced GPLLMs, and our KG-o1 models on four datasets. Afterwards we provide a detailed analysis.

Datasets

We conduct our main experiments on three open-source datasets: HotpotQA, 2WikiMultiHopQA, and MINTQA as well as our self-constructed KG-MHQA test dataset. Dataset statistics are shown in Table 3. For HotpotQA and 2WikiMultiHopQA, we follow the standard experimental setup of previous work (Trivedi et al. 2023), using 500 sampled instances for each dataset. For MINTQA (He et al. 2024), which focuses on new and tail knowledge, we remove questions whose entities overlap with the KG-MHQA SFT training data to prevent knowledge leakage, and then uniformly sample 300 questions requiring 2–4 reasoning hops.

For our self-built KG-MHQA test set, we conduct entity-based partitioning and rigorous manual annotation to ensure data quality, resulting in a test set of 307 high-quality samples. The annotation team, comprising one PhD student and one Master’s student with NLP expertise, independently verified both correctness and reasoning coherence of QA pairs, resolving discrepancies through group discussion. The inter-annotator agreement report is in Section “Inter-annotator agreement report” in the Appendix.

Baseline Models

We adopt the Qwen2.5-14B-Instruct, Qwen2.5-7B-Instruct, Llama3.1-8B-Instruct, and DeepSeek-R1-Distill-Qwen-14B as our backbones. The primary experiments focus on Qwen2.5-14B-Instruct. For comparison, we select several

Datasets	# Hop	# Test Samples	# Avg. Doc. Len.
HotpotQA	2	500	1791.61
2WikiMultiHopQA	2,4	500	984.52
MINTQA	2 to 4	300	1006.33
KG-MHQA	2 to 6	307	1094.45

Table 3: Statistics and description of test datasets.

prominent LRMs, including o1-mini, DeepSeek-R1, QwQ-32B-Preview, Gemini-2.0-Flash-Thinking, and an influential model, Open-o1, in the open source community. Additionally, we include two advanced general-purpose large language models (GPLLMs): ChatGPT-4o and ChatGPT4o-mini.

Implementations and Metrics: In our experiments, we train our KG-o1 models on the KG-MHQA training set through SFT and DPO. Our training details are shown in Appendix Section “Training Details”. For evaluation, we report exact match (EM), F1 score (F1), precision (Prec.), and recall (Rec.) results of baseline models in four datasets. The settings of evaluation are shown in Appendix Section “Evaluation Metrics and Hyper-parameter Setting”.

Main Results

We compare the KG-o1 models with several baselines, including both SFT and preference learning approaches, as well as advanced LRMs, powerful GPLLMs, and basic LLMs. The results are presented in Table 2. Our experiments demonstrate that advanced LRMs perform well in multi-hop question answering, with Deepseek-R1 and Gemini-2.0-Flash-Thinking-exp leading the pack, outperforming the other LRMs by a wide margin. Meanwhile, preference learning methods fur-

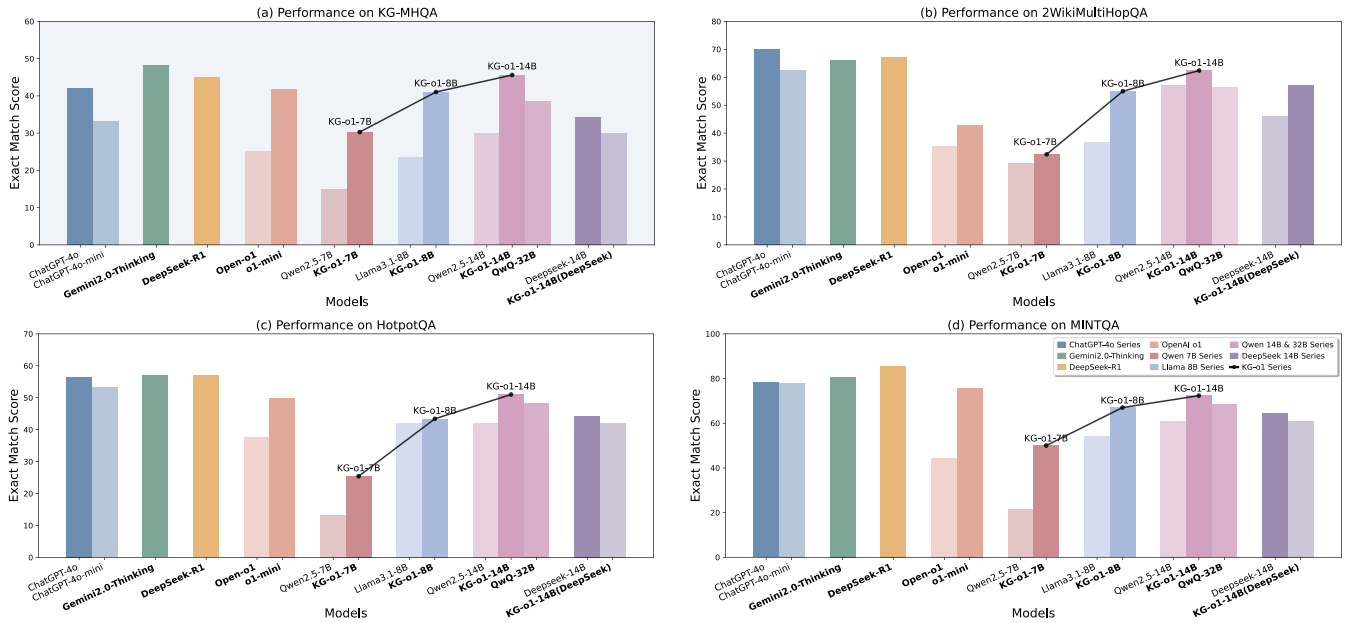


Figure 3: Comparison of model size and performance for GPLLMs, LRMs, and KG-o1 models on four multi-hop reasoning datasets using Exact Match as a metric. Long-term thinking models are indicated in bold.

ther enhance the reasoning capabilities of open-source models, both our KG-o1 and GRPO training method substantially improve the reasoning performance. Notably, GPLLMs achieve strong results without employing the long-term thinking paradigm, even surpassing Deepseek-R1 on 2WikiMultiHopQA, indicating that advanced GPLLMs can excel with traditional CoT. When comparing KG-o1 models with their base models, our method significantly improves performance across the three open-source and KG-MHQA test datasets, highlighting both its effectiveness and generalization ability.

Next, we analyze LLMs’ multi-hop reasoning ability through two dimensions: knowledge ability and reasoning ability. Since this ability cannot be directly decomposed, we link knowledge ability to parameter scale and reasoning ability to reasoning paradigms. Figure 3 visualizes performance variations across model series.

Analyzed from knowledge capability. As model parameters increase, the performance of LLMs within the same series and reasoning paradigm improves significantly. For instance, the performance boost from the 7B to 14B parameter scale in the Qwen series is substantial. However, once the number of parameters reaches a certain threshold, further increases yield diminishing returns in knowledge capacity. For example, on the MINTQA dataset, the performance gain between the 40-mini and 40 models is only 0.33, with a similar trend observed across the other three datasets. This suggests that as model size grows, the marginal improvement in knowledge capacity and reasoning ability declines.

Analyzed from reasoning ability. Comparing LLMs with different inference patterns at the same parameter scale reveals that the long-term thinking model exhibits superior multi-hop reasoning ability. Specifically, experiments comparing our trained KG-o1 model with base models (e.g., KG-

o1-7B vs. Qwen2.5-7B, KG-o1-14B vs. Qwen2.5-14B) show significant performance improvements. To further explore the relationship between reasoning ability and patterns, we conducted ablation studies on four datasets. As shown in (a) in Figure 3, the KG-o1 model outperforms the base model after training, indicating that the long-term thinking paradigm enhances multi-hop reasoning. Notably, even when knowledge ability improvements are similar (as seen from the comparable slopes in (a) in Figure 3), the reasoning ability improvement is substantially greater than in the base model. This finding is further validated across three generalized experimental datasets, where both KG-o1-7B and KG-o1-14B show consistent performance gains, demonstrating the effectiveness of our approach in enhancing multi-hop reasoning.

Ablation Study

In our approach, we systematically conduct ablation experiments for different training phases and reinforcement learning methods, the results are presented in Table 4. In addition, we specifically designed the ablation experiments for medical domain transferability.

SFT LLM (QA) vs. CoT: SFT LLM (QA) is the model training only on the QA dataset. The QA model shows limited improvement or even a decrease in performance compared to the CoT approach, e.g., on the 2Wikimultihopqa dataset, the performance decreases to 36.80%.

SFT LLM (QA) vs. SFT LLM: SFT LLM is trained solely on the KG-MHQA SFT dataset. Comparing the SFT LLM with the QA LLM reveals that while both models incorporate knowledge from the training set, the long-thinking model can effectively analyze the question, plan the solution, integrate knowledge into the reasoning chain, and iteratively validate it to arrive at the final answer.

Method	KG-MHQA †				2WikiMultiHopQA				HotpotQA				MINTQA			
	EM	F1	Prec.	Rec.	EM	F1	Prec.	Rec.	EM	F1	Prec.	Rec.	EM	F1	Prec.	Rec.
CoT	29.97	38.97	37.64	54.78	57.20	68.15	65.86	79.44	42.20	55.31	55.14	78.66	61.00	70.50	68.77	89.18
SFT (QA)	43.65	65.22	60.05	78.42	36.80	54.15	50.83	62.47	42.40	60.19	61.34	64.18	60.00	76.86	72.84	88.58
SFT	41.37	58.89	54.92	69.27	59.20	72.25	70.47	77.45	48.60	66.67	68.85	68.32	68.33	76.89	75.84	80.77
DPO	30.07	39.98	38.38	52.45	46.89	62.91	59.37	74.65	42.37	59.81	60.31	70.00	66.67	78.14	76.62	86.43
GRPO	34.96	45.71	43.91	54.93	51.50	66.47	63.25	76.58	47.59	64.84	65.98	69.36	67.33	78.51	77.35	86.85
SADPO	34.31	45.81	43.79	57.73	45.89	62.87	59.23	74.98	44.18	62.25	62.11	72.50	64.67	75.67	74.60	85.00
SFT + DPO	41.37	59.12	55.14	69.58	61.00	73.09	71.33	78.07	48.80	65.72	67.75	67.53	69.67	76.59	75.69	79.26
SFT + GRPO	43.13	59.62	55.68	69.68	61.01	72.59	70.90	77.29	48.39	65.75	67.72	68.13	71.61	78.13	77.60	80.35
SFT + SADPO	45.60	69.36	63.75	84.46	62.40	74.45	72.56	79.45	51.00	67.11	69.66	68.18	72.33	79.37	78.73	81.49

Table 4: Exact-Match (EM), F1, Precision, and Recall results of ablation study of Qwen2.5-14B-Instruct on multi-hop question answering datasets. † represents our self-build dataset, SADPO denotes Self-improved Adaptive DPO.

RL vs. SFT: Experimental results indicate that reinforcement learning (RL) surpasses supervised fine-tuning (SFT) in boosting LLMs’ multi-hop reasoning. Notably, Self-improved Adaptive DPO outperforms GRPO in guiding LLMs to produce coherent long-term reasoning and accurate answers, which stems from its adaptive self-improvement and preference-adjustment mechanisms.

Domain Transferability: We choose the MedQA-MedGENIE test set (Frisoni et al. 2024) for domain transferability experiment. We use GPT-4o to reformat the multiple-choice questions to generative questions and use the provided contexts as the input knowledge. Experimental result in Table 5 shows that GPLLMs, such as ChatGPT-4o, demonstrate superior performance over LRM on the medical reasoning dataset. Meanwhile, our proposed method (trained solely on a general knowledge graph) significantly improves the medical reasoning ability of the base model, achieving competitive performance comparable to DeepSeek-R1. This validates the effectiveness of our methods in cross-domain transferability.

Model	Method	EM	F1	Prec.	Rec.
ChatGPT-4o-mini	CoT	20.68	38.73	36.81	47.54
ChatGPT-4o	CoT	25.24	44.33	42.78	51.47
Qwen2.5-14B	CoT	13.87	28.62	26.16	46.33
DeepSeek-R1	Reasoning	17.49	27.85	27.64	32.52
DeepSeek-14B	Reasoning	16.65	33.64	30.79	46.62
KG-o1-14B	Reasoning	17.99	32.97	33.66	35.80

Table 5: Performance on MedQA-MedGENIE

Error and Cost Analysis

Figure 4 summarizes both error analysis and resource-performance trade-offs for KG-o1 and baseline models across four datasets. In (a), we manually classify 100 randomly sampled errors into knowledge errors, where the model uses irrelevant or incorrect knowledge, and logic errors, which result from faulty multi-hop reasoning. Results show that our approach substantially reduces knowledge errors on KG-MHQA and logic errors on the other datasets, especially 2WikiMultiHopQA, demonstrating the effectiveness of our approach (see Appendix Section “Cases of Two Error Types”

for details). In (b), as the size of the model parameter increases, the KG-o1 models generate shorter outputs, consume fewer tokens, and achieve better performance, a trend also seen in QWQ-32B-preview and Deepseek-R1. Furthermore, KG-o1 models display more consistent token usage compared to Deepseek-R1, likely because our training set’s long reasoning sequences encourage stable response lengths.

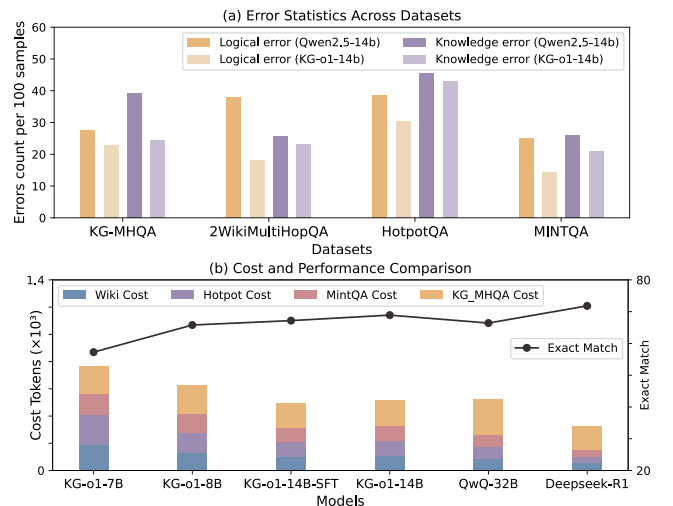


Figure 4: (a) Error statistics for Qwen2.5-14B-Instruct and KG-o1-14B models on four datasets. (b) Cost and overall performance of six LRMs across four datasets

Conclusion and Limitations

In conclusion, we propose KG-o1, which integrates KGs to evolve an LLM to an LRM for complex knowledge-intensive multi-hop question answering. We first select complex fact subgraphs from KGs and generate logical paths. Further, we construct complex multi-hop QA pairs and iteratively generate the KG-MHQA SFT dataset for KG-driven long-term thinking SFT. Additionally, we construct the KG-MHQA test dataset for model evaluation. Finally, we propose the “Self-improved Adaptive DPO” strategy to obtain the adaptive KG-MHQA DPO dataset for the alignment of appropriate

reasoning. We then conducted experiments on four multi-hop reasoning datasets and a domain transferability dataset of advanced LRMs, GPLLMs, and our KG-o1 models, further validating the effectiveness of our proposed method.

Although we have implemented strategies to enhance the quality of the thinking process, our current approach primarily treats the thinking process as a whole for contrastive learning. More granular alignment of the model’s thinking content remains space for future exploration. Furthermore, when constructing multi-hop questions, we focused on free-text answers, overlooking other answer formats such as comparisons or yes/no judgments. These limitations highlight potential directions for improvement, and we will continue to explore ways to refine and advance our work.

Ethical Statement

This paper proposes KG-o1, which integrates KGs to enhance the multi-hop reasoning abilities of LLMs. We ensured ethical compliance by using publicly available or copyright-compliant datasets and avoiding sensitive information. Our model prioritizes factual and logical consistency to reduce misinformation risks, and our methods are transparently documented for reproducibility. We also optimized computational resources to reduce environmental impact, aligning our work with ethical AI research standards. We recognize that applying KG-o1 to reasoning tasks in diverse real-world scenarios may have broader impacts. To prevent potential misuse or misinterpretation, we recommend users validate model outputs with relevant experts before adopting them for important decisions, ensuring responsible and ethical application.

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Access of KG-o1 approach

We provide an anonymous link to a GitHub repository containing **SFT and DPO datasets**, **training hyperparameters**, **evaluation results**, and **KG-o1 models**. Meanwhile, we have added an open-source license to the project.

URL: <https://anonymous.4open.science/r/KG-o1-F493>

Entity-to-Subgraph Expansion Examples.

Step 1: Initial Triplet Units Construction

Formula Definition

$$S_j = \mathbb{U}_{|R|} \{s_r^{e_i}\}, e_i \in N, j = 1 \quad (4)$$

Initial Entity: Batman:_Gotham_Knight

Selected Relation Type:

/film/dubbing_performance/film

Generated Unit:

```
[
  ["Shinichiro_Mik", "/film/dubbing_performance/
  film", "Batman:_Gotham_Knight"],
  ["Kikuko_Inoue", "/film/dubbing_performance/
  film", "Batman:_Gotham_Knight"],
  ["Batman-US", "/film/dubbing_performance/film",
  "Batman:_Gotham_Knight"]
]
```

In this example, N is the set of initial entities, while Batman:_Gotham_Knight represents the explicit entity e_i in N . Select a relation type r connected to e_i , i.e., /film/dubbing_performance/film, to expand the entities (Shinichiro_Mik, Kikuko_Inoue, Batman-US) to get a unit. $\mathbb{U}$ is the set of all units.

Step 2: Triplets Expansion

Formula Definition

$$S_j = S_{j-1} \cup \{(m, r, n_i) \text{ or } (n_i, r, m) | m \notin N, n_i \in N, r \in R\}, j > 1 \quad (5)$$

Selected Relation Type:

/tv/special_tv_performance_type/episode_performances

Expanded Subgraph:

```
[
  ["Shinichiro_Mik", "/film/dubbing_performance/
  film", "Batman:_Gotham_Knight"],
  ["Kikuko_Inoue", "/film/dubbing_performance/
  film", "Batman:_Gotham_Knight"],
  ["Batman-US", "/film/dubbing_performance/film",
  "Batman:_Gotham_Knight"],
  ["Seiyu-GB", "/tv/special_tv_performance_type/
  episode_performances", "Shinichiro_Mik"],
  ["Seiyu-GB", "/tv/special_tv_performance_type/
  episode_performances", "Kikuko_Inoue"]
]
```

R represents the concatenated set of all relational types, each relational type r is connected with one or more nodes in N . In this example, r is /tv/special_tv_performance_type/episode_performances.

Next, we need to find an entity connected to r and contained in N as n_i . Locate the triplets in KG and add them to

the unit. The triplets should satisfy that the relationship type is r and one of the head or tail entities is n_i . For example, two triplets were added in this round of extensions:

```
[
  ["Seiyu-GB", "/tv/special_tv_performance_type/
  episode_performances", "Shinichiro_Mik"],
  ["Seiyu-GB", "/tv/special_tv_performance_type/
  episode_performances", "Kikuko_Inoue"]
]
```

Logical Triplets Generation Examples.

Step 1: Cluster Triplet Formation

Original Triplets:

```
[
  ["Shinichiro_Mik", "/film/dubbing_performance/
  film", "Batman:_Gotham_Knight"],
  ["Kikuko_Inoue", "/film/dubbing_performance/
  film", "Batman:_Gotham_Knight"],
  ["Batman-US", "/film/dubbing_performance/film",
  "Batman:_Gotham_Knight"],
  ["Seiyu-GB", "/tv/special_tv_performance_type/
  episode_performances", "Shinichiro_Mik"],
  ["Seiyu-GB", "/tv/special_tv_performance_type/
  episode_performances", "Kikuko_Inoue"],
  ["Sailor_Moon_S_the_Movie", "/film/performance/
  special_performance_type", "Seiyu-GB"],
  ["Case_Closed:_Captured_in_Her_Eyes", "/film/
  performance/special_performance_type",
  "Seiyu-GB"],
  ["Dragon_Ball:_Mystical_Adventure", "/film/
  performance/special_performance_type",
  "Seiyu-GB"]
]
```

Triplets After Entity Clustering:

```
[
  ["Shinichiro_Mik", "Kikuko_Inoue"], "/film/
  dubbing_performance/film",
  "Batman:_Gotham_Knight"],
  ["Batman-US", "/film/dubbing_performance/film",
  "Batman:_Gotham_Knight"],
  ["Seiyu-GB", "/tv/special_tv_performance_type/
  episode_performances", ["Shinichiro_Mik",
  "Kikuko_Inoue"]],
  ["Sailor_Moon_S_the_Movie", "Case_Closed:
  _Captured_in_Her_Eyes", "Dragon_Ball:
  _Mystical_Adventure"], "/film/performance/
  special_performance_type", "Seiyu-GB"]
]
```

We group triplets sharing the *same relation and connecting entity*. For instance, original triplets with /film/dubbing_performance/film and tail Batman:_Gotham_Knight:

- [Shinichiro_Mik, Kikuko_Inoue] $\rightarrow$ film $\rightarrow$ Batman (clustered head)
- Batman-US $\rightarrow$ film $\rightarrow$ Batman (single head preserved)

This preserves semantic equivalence while reducing redundancy.

Step 2: Logical Triplet Conversion

Triplets After Entity Clustering:

```
[
  ["Shinichiro_Mik", "Kikuko_Inoue"], "/film/
  dubbing_performance/film",
  "Batman:_Gotham_Knight"],
  ["Batman-US", "/film/dubbing_performance/film",
  "Batman:_Gotham_Knight"],
  ["Seiyū-GB", "/tv/special_tv_performance_type/
  episode_performances", ["Shinichiro_Mik",
  "Kikuko_Inoue"]],
  ["Sailor_Moon_S_the_Movie", "Case_Closed:
  _Captured_in_Her_Eyes", "Dragon_Ball:
  _Mystical_Adventure"], "/film/performance/
  special_performance_type", "Seiyū-GB"]
]
```

Logical Triplets:

```
[
  ["#0", "/film/dubbing_performance/film", "#1"],
  ["Batman-US", "/film/dubbing_performance/film",
  "#1"],
  ["#2", "/tv/special_tv_performance_type/
  episode_performances", "#0"],
  ["Sailor_Moon_S_the_Movie", "Case_Closed:
  _Captured_in_Her_Eyes", "Dragon_Ball:
  _Mystical_Adventure"], "/film/performance/
  special_performance_type", "#2"]
]
```

We replace concrete entities with logical placeholders (#0, #1, etc.) to create abstract patterns:

- Clustered head [Shinichiro_Mik, Kikuko_Inoue] becomes #0
- Batman:_Gotham_Knight becomes #1

Thus, [#0, film, #1] and [Batman-US, film, #1] logically represent both clustered and individual relationships.

Interrogative Sentence Syntax

Interrogative sentences consist of three key components: **the subject of the question, bridge entities, and sources providing factual information**. These components are essential for constructing meaningful and logically coherent multi-hop reasoning tasks, which require the ability to traverse and link disparate pieces of information across multiple steps.

For example, a typical multi-hop reasoning question could be: “Her birth year is 1997, and her husband’s father is the founder of company 1. Who is she?” In this question, the subject of the inquiry is “her”, and the associated modifier components include: “her birth year is 1997” and “her husband’s father is the founder of company 1”. In this example, “her husband’s father” serves as a bridge entity, linking “her husband” with “the founder of company 1.” At the same time, “Company 1” and “1997” function as sources providing factual information, which further restrict the potential answers and help clarify the subject being inquired about. In this example, “her husband’s father” acts as a bridge entity, connecting “her husband” with “the founder of company 1”

Meanwhile, “Company 1” and “1997” function as sources providing factual information that help to narrow down the possible answers. The correct answer hinges on correctly interpreting the syntactic structure of the question, linking the entities and pieces of information in a logical sequence.

Inter-annotator agreement report

(1) Independent Annotation Stage: Two graduate school students (A&B) conducted blind verification on 387 raw QA pairs using predefined criteria. Researcher A retained 303 samples (78.3% retention), while Researcher B approved 311 samples (80.3% retention).

(2) Consensus-building Stage: For the 26 non-overlapping samples (A:5 unique, B:8 unique, plus 13 divergently judged cases), we conducted joint review sessions through completeness analysis and error pattern discussion (e.g., ambiguous questions or wrong answers), 9 additional high-quality samples were validated.

The Kappa coefficient is 0.795, indicating high agreement between the two labelers in the independent annotation phase.

Evaluation Metrics and Hyper-parameter Setting

Generally, We use Exact Match (EM), F1-score, Precision, and Recall to evaluate the performance of models on multi-hop reasoning tasks. EM measures the percentage of predictions that exactly match the ground truth. F1-score provides a balanced measure between Precision and Recall. Precision evaluates the accuracy of the models’ predicted answers, while Recall assesses the models’ ability to capture relevant answers. These metrics provide a comprehensive evaluation of the performance of LLMs on multi-hop reasoning tasks.

For evaluation, we employed the VLLM toolkit (Kwon et al. 2023) for accelerated reasoning, setting the temperature to 0.5 and top_p to 0.9. For the LRMs and KG-o1 models, the maximum token limit was configured to 8,192, while models without long-thinking capabilities had a token limit of 1,024.

Training Details

Supervised fine-tuning for the models Qwen2.5-14B-Instruct, Qwen2.5-7B-Instruct, Llama3.1-8B-Instruct and DeepSeek-R1-Distill-Qwen-14B was conducted using Llama-Factory (Zheng et al. 2024) across four NVIDIA A8000 80GB GPUs. The training hyperparameters were set as follows: learning rate as 1.0e-4 and batch size as 16. For direct preference optimization, the same hardware configuration was employed, with the hyperparameters adjusted to: learning rate as 1.0e-5 and batch size as 4. Efficient training was ensured throughout DeepSpeed ZeRO-3 optimization (Rasley et al. 2020).

Dataset Curation Prompts

Figure C1 illustrates the prompt used for question generation

Figure C2 illustrates the prompt used for question Verification

Figure C3 illustrates the prompt used for thinking distillation

Figure C4 illustrates the prompt used for KG-driven Thought Optimization

Figure C5 illustrates the prompt used for KG-driven Thought Optimization in the final step.

Figure C6 illustrates the prompt used for Output Generation.

Prompt for Question Generation:

Please generate a multi-hop reasoning question based on the triplet information ,the specified question subject and , with the following requirements:

1. Identifiers in the question must appear as pronouns (e.g., “she,” “he”) or as “pronoun + relationship” (e.g., “his wife”) rather than using node names. Moreover, the pronoun form of the questioned entity should be determined by the answer.
2. The resulting multi-hop problem must contain all the information from the triplet, especially the non-Identifiers nodes in the triplet.
3. Placeholders like "#0" or other abstract tags can't be included in the question.
4. Ensure that the generated question is logically clear and semantically fluent.
5. Ensure that the answer does not appear in the question.

##Examples##

{Examples}

##Triplet Information##

Questioned_Entity: {q entity}

Triplet_Information: {logic subgraph}

Answer: {entity}

Please output in the following JSON format, Ensure that ...

```
```json
{
 "MHQuestion": xxx,
 "Questioned_Entity": xxx
}
```
```

Figure C1: Prompt for Question Generation.

Prompt for Quesiton Decomposition:

Please analyze the following multi-step reasoning question and break it down into multiple sub-problems. Each sub-problem must correspond to one reasoning triplet from the provided reasoning path. Ensure the number of sub-problems matches the number of reasoning triplets, and there is a one-to-one correspondence between them.

For each sub-problem:

1. Provide a clear and concise description of the sub-problem.
2. Sub-problem should not contain placeholders like "#0" or other abstract labels.
3. Explain the reasoning and thought process for breaking down the original question into this specific sub-problem.
4. Assign the corresponding reasoning triplet from the provided reasoning path.

Multi-step Reasoning Problem

{q}

Reasoning Path

{logic_sg}

Output Requirements

Ensure that the JSON output ... Ensure that the "reasoning triplet" field in each sub-problem is selected directly from the reasoning path provided:

```
```json
[
 {
 "sub_problem_1": "xxx",
 "reason": "xxx",
 "reasoning_triplet": "xxx"
 },
 {
 "sub_problem_2": "xxx",
 "reason": "xxx",
 "reasoning_triplet": "xxx"
 }
]
```
```

Figure C2: Prompt for Question Verification

Prompt for Thinking Distillation:

I will provide you with a multi-hop question, along with its corresponding decomposed sub-questions and reasoning steps. Based on the given information, your task is to solve the multi-hop question based on its corresponding decomposed sub-questions. Finally, provide the answer to the multi-hop question clearly and concisely.

Make sure to:

The overall solution to the problem can refer to the decomposed sub-problems and make appropriate supplements.

Question: {*question*}

Corresponding decomposed sub-questions:

```
{{
    'sub_problem_1': xxx,
    'sub_problem_2': xxx,
    ...
}}
```

Figure C3: Prompt for Thinking Distillation

Prompt for KG-driven Thought Optimization:

Multi-hop Question: $\{q\}$

Corresponding Sub-questions: $\{sub_q\}$

The brainstorming_process for solving $\{sub_question\}$ is as follows: $\{thinking_process\}$

Knowledge Source: $\{knowledge\}$

Answer of the sub-question: $\{sub_answer\}$

Based on the original writing style of brainstorming_process (i.e., title + discussion text) and the logical connections between the titles, use the knowledge provided to generate a brainstorming process and candidate answers for addressing the sub-question. The requirements are as follows:

1. Follow the format of the given brainstorming_process, including title and discussion text.
2. Make the brainstorming_process more comprehensive by discussing background information, such as the characteristics and attributes of relevant options, as well as different categories of institutions or organizations.
3. Utilize all the provided knowledge in the generated text.
4. Ensure the generated brainstorming_process is detailed.
5. The number of (title, discussion_text) segments in the generated brainstorming_process should be significantly more than those in the given brainstorming_process.
6. At the part of candidate_answers, you need to provide several candidate answers in natural language, and these answers must include the given real answer of the sub-question.

Output the final brainstorming process in JSON format.

```
```json
{{
 "brainstorming_process": "An unstructured natural language text",
 "candidate_answers": "several candidate answers in natural language"
}}
```
```

Figure C4: Prompt for KG-driven Thought Optimization.

Prompt for Last Step KG-driven Thought Optimization:

Multi-hop Question: $\{q\}$

Corresponding Sub-questions: $\{\textit{sub } q\}$

The brainstorming_process for solving $\{\textit{sub question}\}$ is as follows: $\{\textit{thinking process}\}$

Knowledge Source: $\{\textit{knowledge}\}$

Answer of the Multi-hop Question: $\{\textit{ground truth}\}$

Taking into account your candidate_answers and brainstorming_process of the previous sub-questions, complete the following tasks:

Based on the original brainstorming_process's writing style (i.e., title + discussion text) and the logical connections between the titles, use the knowledge provided to generate a brainstorming process for addressing the sub-question, generate the final answer to the Multi-hop Question. The requirements are as follows:

1. Follow the format of the given brainstorming_process, including title + discussion text. The number of title + discussion text segments should exceed those in the original text.
2. Make the brainstorming_process more comprehensive by discussing background information, such as the characteristics and attributes of relevant options, as well as different categories of institutions or organizations.
3. Utilize all the provided knowledge in the generated text.
4. Ensure the generated brainstorming_process is detailed.
5. The number of title + discussion text segments in the generated brainstorming_process should be significantly more than those in the given brainstorming_process.
6. At the final_answer part, your answer should be the given final answer to the multi-hop problem.

Output the final brainstorming process in JSON format.

```
```json
{{
 "brainstorming_process": "An unstructured natural language text",
 "final_answer": "the final answer"
}}
```

Figure C5: Prompt for Last Step KG-driven Thought Optimization.

## Prompt for Output Generation:

### Task Description:

Given an input, including a multi-hop question, the process of solving the question, and the correct answer, your task is to generate a process for answering the multi-hop question based on the information provided. The process should include the solution process of the sub-questions and the final answer.

### Input:

Multi-hop Question:  $\{q\}$

Solution Process:  $\{\textit{slow thinking process}\}$

Sub\_questions:  $\{\textit{sub questions}\}$

Correct Answer:  $\{\textit{ground truth}\}$

Output Format:

### Output in JSON format..

```
```json
{{
  "Reasoning_Process": "Briefly describe the reasoning logic and key steps, and provide the final answer"
}}
```

Figure C6: Prompt for Output Generation.

Case Study

Figure D1 to Figure D3 presents the example response of KG-o1-14B on 2WikiMultiHopQA.

Figure D4 to Figure D6 presents the example response of KG-o1-14B on HotpotQA.

Figure D7 to Figure D9 presents the example response of KG-o1-14B on MINTQA.

Figure D10 to Figure D15 presents the example response of KG-o1-14B on KG-MHQA.

Case Study for 2WikiMultiHopQA:

Question: Which one was established first, Grouplogic or Inbios?

Ground Truth: GroupLogic

KG-o1-14B:

<think>

Understanding the Context of GroupLogic: GroupLogic is a well-established enterprise software company that specializes in data management solutions. It was founded in 1988, which places it in the early days of the personal computer revolution. This era was marked by rapid advancements in technology and the growing need for efficient data management systems. GroupLogic's products, such as activeEcho, mobilEcho, and ArchiveConnect, cater to the needs of businesses looking to streamline their data storage and access processes.

Exploring the Origins of InBios: InBios International, Inc., on the other hand, is a medical diagnostic company that focuses on the development of rapid and ELISA-based immunodiagnostic assays. Founded in 1996, InBios is relatively newer compared to GroupLogic. The company's establishment coincided with the increasing global awareness of infectious diseases and the need for rapid diagnostic tools. InBios has developed tests for various diseases, including dengue fever, malaria, and leishmaniasis, which are crucial for public health management.

Comparing the Founding Years: To determine which company was established first, we need to compare the founding years of GroupLogic and InBios. GroupLogic was founded in 1988, while InBios was established in 1996. This indicates that GroupLogic predates InBios by 8 years.

Analyzing the Impact of Founding Years: The difference in founding years between GroupLogic and InBios reflects the technological and medical advancements of their respective eras. GroupLogic's establishment in 1988 aligns with the early stages of the personal computer boom, where businesses were beginning to rely heavily on digital data management. In contrast, InBios's founding in 1996 coincides with the growing global health concerns and the need for rapid diagnostic solutions.

...

Figure D1: Case Study for 2WikiMultiHopQA (1/3)

Case Study for 2WikiMultiHopQA:

Understanding the Significance of Founding Years: The founding years of these companies not only indicate their age but also reflect the technological and medical landscapes of their times. GroupLogic's early establishment in 1988 positioned it to capitalize on the burgeoning enterprise software market, while InBios's later establishment in 1996 allowed it to address the increasing demand for medical diagnostics in the face of global health challenges.

Conclusion: Based on the information provided, GroupLogic was established first, preceding InBios by 8 years. This difference in founding years highlights the distinct technological and medical contexts in which these companies were founded.

Understanding the Context of GroupLogic and InBios: GroupLogic is an enterprise software company that specializes in data management solutions, particularly focusing on file sharing and collaboration tools. It was established in 1988, which places it in the early days of the personal computer revolution, a time when businesses were increasingly looking for ways to manage and share data efficiently. InBios, on the other hand, is a medical diagnostic company that develops rapid and ELISA-based immunodiagnostic assays. It was founded in 1996, a period marked by significant advancements in medical technology and the growing need for rapid diagnostic tools to address global health challenges.

Exploring the Founding Years: To determine which company was established first, we need to compare their founding years. GroupLogic was founded in 1988, while InBios was established in 1996. This indicates that GroupLogic predates InBios by 8 years.

Analyzing the Impact of Founding Years: The difference in founding years between GroupLogic and InBios reflects the technological and medical advancements of their respective eras. GroupLogic's establishment in 1988 aligns with the early stages of the personal computer boom, where businesses were beginning to rely heavily on digital data management. In contrast, InBios's founding in 1996 coincides with the growing global health concerns and the need for rapid diagnostic solutions.

Understanding the Significance of Founding Years: The founding years of these companies not only indicate their age but also reflect the technological and medical contexts in which they were founded. GroupLogic's early establishment in 1988 positioned it to capitalize on the burgeoning enterprise software market, while InBios's later establishment in 1996 allowed it to address the increasing demand for medical diagnostics in the face of global health challenges.

...

Figure D2: Case Study for 2WikiMultiHopQA (2/3)

Case Study for 2WikiMultiHopQA :

Conclusion: Based on the information provided, GroupLogic was established first, preceding InBios by 8 years. This difference in founding years highlights the distinct technological and medical contexts in which these companies were founded.

</think>

<output>

To determine which company was established first between GroupLogic and InBios, we need to compare their founding years. GroupLogic was founded in 1988, while InBios was established in 1996. By comparing these years, we can conclude that GroupLogic predates InBios by 8 years. Therefore, GroupLogic was established first. The final answer is GroupLogic.

The final answer is: GroupLogic

</output>

Figure D3: Case Study for 2WikiMultiHopQA (3/3)

Case Study for HotpotQA:

Question:

Are Catasetum and Origanum in the same family?

Ground Truth: no

KG-o1-14B:

<think>

Understanding the Family of Catasetum: Catasetum is a genus of orchids, which are known for their unique and often showy flowers. Orchids belong to the family Orchidaceae, which is one of the largest families of flowering plants. This family is characterized by its diverse range of species, many of which are epiphytic, meaning they grow on other plants without being parasitic. The Orchidaceae family is known for its complex reproductive mechanisms and often intricate flower structures.

Characteristics of the Orchidaceae Family: The Orchidaceae family is distinguished by several key characteristics. Orchids typically have bilateral symmetry in their flowers, meaning they can be divided into two mirror-image halves. They often have a specialized pollination mechanism, which can involve various animals, including insects and birds. The family is also known for its wide range of habitats, from tropical rainforests to arctic regions.

Catasetum's Place in the Orchidaceae Family: Catasetum, as a genus within the Orchidaceae family, shares these characteristics with other orchids. It is part of the subfamily Epidendroideae, which is known for its epiphytic growth habit. This subfamily includes many genera that are prized for their ornamental value in horticulture.

Exploring the Lamiaceae Family: In contrast to the Orchidaceae family, the Lamiaceae family, also known as the mint family, includes plants like oregano. This family is characterized by its square stems, opposite leaves, and often aromatic foliage. The Lamiaceae family is known for its culinary and medicinal uses, with many species being used in cooking and traditional medicine.

...

Figure D4: Case Study for HotpotQA (1/3)

Case Study for HotpotQA:

Distinguishing Between Orchidaceae and Lamiaceae: The key differences between the Orchidaceae and Lamiaceae families lie in their floral structures and growth habits. While Orchidaceae plants are often epiphytic and have complex flowers, Lamiaceae plants are typically terrestrial and have simpler, more uniform flowers. These differences help in identifying the family to which a plant belongs.

Conclusion: Based on the characteristics and family traits discussed, it is clear that *Catasetum* belongs to the Orchidaceae family, which is distinct from the Lamiaceae family where oregano is found. This distinction is crucial for understanding the evolutionary and ecological relationships between these plants.

Understanding the Family of *Origanum*: *Origanum* is a genus of plants that belongs to the mint family, Lamiaceae. This family is known for its aromatic herbs and is characterized by square stems, opposite leaves, and a strong scent. *Origanum* includes several species, such as *Origanum vulgare* (common oregano), which is widely used in culinary applications. The Lamiaceae family is diverse, containing many economically important plants like mint, basil, and sage.

Exploring the Characteristics of Lamiaceae: The Lamiaceae family, also known as the mint family, is a large family of flowering plants. It is characterized by its aromatic foliage, square stems, and opposite leaves. This family includes a wide variety of plants used in both culinary and medicinal applications. The strong scent and flavor of many Lamiaceae species make them popular in cooking and herbal remedies.

Comparing *Catasetum* and *Origanum*: *Catasetum* is a genus of orchids, belonging to the Orchidaceae family. This family is known for its complex reproductive mechanisms and diverse range of species. Unlike the Lamiaceae family, Orchidaceae plants are often epiphytic or terrestrial and have intricate flower structures. The key difference between *Catasetum* and *Origanum* lies in their family classification, with *Catasetum* being part of the Orchidaceae and *Origanum* part of the Lamiaceae.

Distinguishing Between Orchidaceae and Lamiaceae: The Orchidaceae family is distinguished by its unique floral structures and epiphytic growth habits, while the Lamiaceae family is recognized by its square stems and aromatic properties. These differences help in identifying the family to which a plant belongs. *Catasetum*, being an orchid, falls under the Orchidaceae family, whereas *Origanum*, as a mint, is part of the Lamiaceae family.

...

Figure D5: Case Study for HotpotQA (2/3)

Case Study for HotpotQA:

Conclusion: Based on the family classification, Catasetum and Origanum are not in the same family. Catasetum belongs to the Orchidaceae family, while Origanum is part of the Lamiaceae family. This distinction is important for understanding the evolutionary and ecological relationships between these plants.

</think>

<output>

To determine if Catasetum and Origanum are in the same family, we need to identify the family of each genus. Catasetum is a genus of orchids, which belong to the Orchidaceae family. This family is known for its complex floral structures and epiphytic growth habits. On the other hand, Origanum is a genus of plants that belongs to the Lamiaceae family, also known as the mint family. This family is characterized by its aromatic foliage, square stems, and opposite leaves. Since Catasetum and Origanum belong to different families, the answer to the question is that they are not in the same family.

The final answer is: No

</output>

Figure D6: Case Study for HotpotQA (3/3)

Case Study for MINTQA:

Question:

Who is the director of the religion or worldview followed by Anas Khalifa?

Ground Truth: Jibril

KG-o1-14B:

<think>

Understanding Anas Khalifa's Religious Affiliation: To determine the religion or worldview followed by Anas Khalifa, we first need to explore his background and any publicly available information about his religious beliefs. Anas Khalifa is a male individual, and understanding his cultural and social context can provide insights into his religious practices.

Exploring Islam as a Major Religion: Islam is one of the world's major religions, characterized by its emphasis on the Five Pillars of Islam, which include Shahada (declaration of faith), Salah (prayer), Zakat (charity), Sawm (fasting during Ramadan), and Hajj (pilgrimage to Mecca). It is practiced by Muslims worldwide and is based on the teachings of the Qur'an, which is considered the holy book of Islam.

Islam's Cultural and Historical Context: Islam originated in Mecca and is centered around the Kaaba, a sacred site in the city. It is distinct from other major world religions such as Christianity, Judaism, and Hinduism. The religion emphasizes concepts like obedience, allegiance, and devotion, and it is known for its rich cultural and historical significance.

Islam's Global Presence: Islam is practiced in various countries, including the United Arab Emirates, Qatar, and the Maldives. It is a religion that spans multiple languages and cultures, reflecting its global reach and influence. Understanding the geographical spread of Islam can help in identifying its followers, including individuals like Anas Khalifa.

Comparing Islam with Other Religions: Islam is often compared with other religions like Christianity and Judaism, which share some historical and cultural roots but differ in their beliefs and practices. This comparison can help in distinguishing the unique characteristics of Islam and identifying its adherents.

Identifying Anas Khalifa's Religion: Given the information that Anas Khalifa is a male individual, and considering the prevalence of Islam in his region, it is likely that his religion is Islam. This conclusion is based on the cultural and religious context of the areas where Islam is predominantly practiced ...

Figure D7: Case Study for MINTQA (1/3)

Case Study for MINTQA:

Conclusion: Based on the analysis of Anas Khalifa's background and the characteristics of Islam, it is reasonable to conclude that his religion or worldview is Islam. This conclusion is supported by the cultural and religious context of the regions where Islam is widely practiced.

Understanding the Concept of a Director in Religion: In the context of religions, the term 'director' can be interpreted in various ways. Typically, it might refer to a leader or a guiding figure within a religious organization or community. However, in the case of Islam, the concept of a 'director' is not as straightforward as in other religions or organizations.

Exploring the Role of Jibril in Islam: In Islam, Jibril, also known as Gabriel, is a significant figure. He is considered one of the archangels and is believed to be the messenger of Allah. Jibril plays a crucial role in conveying divine messages to prophets, including Muhammad, the founder of Islam. This role makes Jibril a key figure in Islamic teachings and practices.

The Role of the Prophet Muhammad: The Prophet Muhammad is central to Islam. He is considered the last prophet and the founder of the religion. His teachings and life are the foundation of Islamic beliefs and practices. While Muhammad is not referred to as a 'director' in the traditional sense, his role as a leader and guide is pivotal in Islam.

Comparing Religious Leadership Structures: Different religions have different structures for leadership. In Christianity, for example, the Pope is the head of the Catholic Church, and in Judaism, the role of a rabbi is significant. In Islam, the concept of leadership is more complex, involving both spiritual guidance and community leadership.

The Significance of Jibril in Islamic Beliefs: Jibril's role as a messenger is crucial in Islamic belief. He is responsible for delivering the revelations of the Qur'an to Muhammad, which are considered the word of Allah. This makes Jibril a central figure in the Islamic worldview, although not in the same way as a human leader.

Understanding the Islamic Worldview: Islam is a monotheistic religion that emphasizes the belief in one God, Allah. It is based on the teachings of the Qur'an and the Hadith, which are the sayings and actions of the Prophet Muhammad. The religion is practiced by Muslims worldwide and is characterized by its emphasis on submission to the will of Allah.

...

Figure D8: Case Study for MINTQA (2/3)

Case Study for MINTQA:

The Role of Angels in Islam: In Islam, angels play a significant role. They are believed to be spiritual beings that carry out the will of Allah. Jibril, as an angel, is particularly important due to his role in conveying divine messages. This role is unique and sets him apart from other figures in Islamic belief.

Conclusion: Based on the information provided and the understanding of Islamic beliefs, Jibril is the figure that can be considered the 'director' of Islam in the context of this question. His role as the messenger of Allah and his significance in delivering the revelations of the Qur'an make him a central figure in the Islamic worldview.

</think>

<output>

To answer the multi-hop question, we need to break it down into two sub-questions. First, we identify the religion or worldview followed by Anas Khalifa. Given the context and cultural background, it is likely that Anas Khalifa follows Islam. Once we establish that Anas Khalifa's religion is Islam, the next step is to determine who the 'director' of Islam is. In Islamic belief, Jibril (Gabriel) is a significant figure who acts as the messenger of Allah, delivering divine messages to prophets, including Muhammad. Therefore, Jibril is considered the 'director' of Islam in the context of this question.

The final answer is: Jibril

</output>

Figure D9: Case Study for MINTQA (3/3)

Case Study for KG-MHQA:

Question:

He is a musician whose place of origin is also tied to a site where both William Powell and Richard Thorpe passed away. Additionally, he was nominated for an award in music composition for *The Quiet Man* and another unspecified nomination. Who is he?

Ground Truth: Victor_Young

KG-o1-14B:

<think>

Understanding the Context of Death Locations:

To determine the place of death for both William Powell and Richard Thorpe, it's important to first understand the context of these individuals. William Powell was an American actor, known for his work in classic Hollywood films, while Richard Thorpe was a film director. Both were prominent figures during the Golden Age of Hollywood.

Exploring the Hollywood Connection:

Hollywood, located in Los Angeles, California, is a significant hub for the film industry and has been the place of death for many notable figures in film and entertainment. This makes it a likely location for both Powell and Thorpe.

Hollywood's Influence on Hollywood-Related Deaths:

The city of Los Angeles, home to Hollywood, has been the final resting place of numerous actors and directors due to its central role in the entertainment industry. This includes individuals like Michael Clarke Duncan, Sammy Cahn, and Joseph Ruttenberg, who all passed away in Los Angeles. Los Angeles as a Common Denominator: Given the prominence of Los Angeles in the entertainment world, it's reasonable to assume that both William Powell and Richard Thorpe, being influential figures in the film industry, would have strong ties to Los Angeles. Therefore, Los Angeles is a plausible location for their deaths.

Exploring Other Notable Deaths in Los Angeles:

Los Angeles has been the site of the deaths of many other significant figures beyond Powell and Thorpe. This includes actors such as Basil Rathbone, Anne Bancroft, and Murray Rothbard. This pattern further supports the likelihood of Powell and Thorpe sharing this location as their place of death. Investigating Other Locations: While exploring potential locations, it's important to consider other notable death sites such as Beverly Hills, Miami Beach, and New York City, each known for their connections to prominent individuals. However, Los Angeles remains the most probable location for Powell and Thorpe due to their industry connections...

Figure D10: Case Study for KG-MHQA (1/6)

Case Study for KG-MHQA:

Verifying the Information:

To confirm the place of death, it is important to cross-reference historical records and biographies of William Powell and Richard Thorpe, which consistently point to Los Angeles as their final place of residence and death. Conclusion: Based on the information and the historical context of Hollywood and Los Angeles, it is highly likely that both William Powell and Richard Thorpe passed away in Los Angeles, California.

Identifying the Place of Death for William Powell and Richard Thorpe: William Powell, a well-known actor, and Richard Thorpe, a film director, both passed away in Palm Springs, California. This location is a notable site for several other celebrities, including Frank Sinatra and Dean Martin. Palm Springs is known for its luxurious resorts and is a popular destination for celebrities, adding to its reputation as a place of rest for many in the entertainment industry.

Exploring the Characteristics of Palm Springs: Palm Springs is a city located in Riverside County, California, known for its desert climate, luxury resorts, and golf courses. It has attracted many celebrities and wealthy individuals, making it a prime location for retirement and relaxation. The city is also recognized for its mid-century modern architecture, which is a significant part of its cultural heritage.

Understanding the Role of Palm Springs in Entertainment History: Palm Springs has been a significant location for many in the entertainment industry. Besides Powell and Thorpe, other notable figures such as Elvis Presley and Bob Hope have also spent time in Palm Springs. This city's appeal to the entertainment elite is due to its beautiful scenery, privacy, and the ability to escape the hustle and bustle of Hollywood.

Associating Musicians with Palm Springs: To find a musician whose place of origin is Palm Springs, we should consider artists who have connections to this location through their career or personal life. This includes exploring musicians who have either been born there or have lived there for a significant period.

The Significance of Palm Springs in Music: While Palm Springs is not traditionally known as a music hub like Los Angeles or Nashville, it has been a retreat for many musicians, providing them with a place to relax and find inspiration. The city's unique environment and culture have influenced various artists over the years.

...

Figure D11: Case Study for KG-MHQA (2/6)

Case Study for KG-MHQA:

Identifying Notable Musicians from Palm Springs: By examining the list of musicians, we find that Victor Young, a renowned composer and conductor, is associated with Palm Springs. Young's career in music composition and his connection to Palm Springs make him a prime candidate for this query.

Cross-referencing with Music Composition Awards: To further validate our findings, we should examine any award nominations for music composition linked to The Quiet Man. Victor Young was indeed nominated for his work on The Quiet Man, which confirms his eligibility as the musician in question.

Understanding the Context of the Question: To solve the problem, we need to identify a musician whose place of origin is linked to a site where both William Powell and Richard Thorpe died. Additionally, this musician should have been nominated for an award in music composition for The Quiet Man and another unspecified nomination. This task requires us to cross-reference various data points about musicians and their origins, as well as links to notable sites of death.

Identifying the Place of Death for William Powell and Richard Thorpe: William Powell and Richard Thorpe both passed away in Palm Springs, California. This information is crucial as it sets the geographical context for the musician's place of origin. Palm Springs is a city known for its luxurious resorts and as a favorite retreat for celebrities, making it a significant site in entertainment history.

Determining the Place of Origin for Musicians: With Palm Springs identified as the site of death for both William Powell and Richard Thorpe, we need to find a musician who was born there. This involves examining the origins of various musicians to see if any have ties to Palm Springs.

Researching Musicians Linked to Palm Springs: After reviewing the available data, Victor Young is noted as originating from Palm Springs. His connection to this location aligns with the sub-question's requirement of identifying a musician whose place of origin is tied to the site of death for William Powell and Richard Thorpe.

Victor Young's Contributions to Music: Victor Young was a renowned film composer and songwriter. His work spans several decades, and he is particularly known for his contributions to film music. This background information makes him a strong candidate for having been nominated for music composition awards...

Figure D12: Case Study for KG-MHQA (3/6)

Case Study for KG-MHQA:

Award Nominations and Recognition: Victor Young was indeed nominated for an Academy Award for his work on 'The Quiet Man.' This nomination is a key point of interest as it matches the criteria outlined in the question. Additionally, further research into his career could reveal other unspecified nominations, reinforcing his standing as a distinguished composer.

Verification of Award Nominations: By verifying Victor Young's nominations for 'The Quiet Man' and potentially others, we can confirm his eligibility as the musician in question. His nomination for 'The Quiet Man' is a clear match, and exploring his full body of work could uncover additional nominations, aligning with the question's requirements.

Conclusion: Based on the gathered information, Victor Young emerges as the musician who fits all the criteria outlined in the question. His origin in Palm Springs, the site of death for William Powell and Richard Thorpe, combined with his nomination for 'The Quiet Man,' makes him the correct answer.

Exploring Award Nominations in Music Composition: To determine if the musician was nominated for an award in music composition for 'The Quiet Man,' we must first delve into the broader landscape of award nominations for film scores. The Golden Globe Award for Best Original Score, for instance, has recognized a wide array of composers, including Philip Glass for 'The Painted Veil' and 'The Little Mermaid,' and John Williams for 'The Aviator.' These nominations highlight the prestige associated with original music compositions in film.

Understanding the Context of 'The Quiet Man': 'The Quiet Man,' a film from the early 20th century, has been recognized with nominations at the Academy Awards. The film's music composition is a critical component, and nominations in this category are indicative of the composer's talent and influence in the film industry. Victor Young, a renowned composer known for his work on classic films, was indeed nominated for the Academy Award for Best Original Score for 'The Quiet Man.' This nomination underscores his significant contribution to the film's success and his reputation in the industry.

Comparing Award Types: It is essential to differentiate between various award nominations. While the Golden Globe Awards recognize achievements in film and television, the Academy Awards focus specifically on achievements in cinematic arts. Victor Young's nomination for an Academy Award in music composition for 'The Quiet Man' places him in the company of other distinguished composers such as Sammy Cahn and Merian C. Cooper, who have been acknowledged for their contributions to the film's success. This nomination highlights Young's expertise and the film's artistic merit...

Figure D13: Case Study for KG-MHQA (4/6)

Case Study for KG-MHQA:

Exploring Other Nominations: In addition to his nomination for 'The Quiet Man,' Victor Young's career includes other significant nominations and awards. His contributions to film music have earned him recognition at various prestigious events, showcasing his versatility and impact on the industry. Understanding these additional nominations helps in comprehensively assessing his career achievements and influence on film music composition.

Analyzing the Role of Film Composers: Film composers like Victor Young play a crucial role in enhancing the narrative and emotional depth of films. Their work is often a reflection of their creative vision and technical skill, and nominations for awards such as the Academy Award for Best Original Score are a testament to their contribution to the art of filmmaking. By examining these nominations, we gain insight into the artistic landscape of film music and the impact of individual composers like Victor Young.

Understanding the Context of the Question: To solve this multi-hop question, we must first understand the key elements involved. We are looking for a musician whose place of origin is linked to a site where both William Powell and Richard Thorpe passed away. Additionally, this musician must have been nominated for an award in music composition for 'The Quiet Man' and another unspecified nomination.

Identifying the Place of Death for William Powell and Richard Thorpe: William Powell and Richard Thorpe both passed away in Palm Springs, California. This information is crucial as it points us to the place of origin of the musician we are trying to identify.

Exploring Musicians Originating from Palm Springs: With Palm Springs identified as the site of death for both William Powell and Richard Thorpe, we need to find a musician from this region. Victor Young, a renowned composer, was born in Palm Springs.

Investigating Award Nominations for Victor Young: Victor Young was indeed nominated for an Academy Award for Best Original Score for 'The Quiet Man.' This matches one of the criteria provided in the question.

Additional Award Nominations for Victor Young: Victor Young's career in music composition was significant, and he received multiple nominations throughout his career. However, the specific details of his other nominations are not provided in the data, but his reputation as a prolific composer suggests he had other notable nominations.

...

Figure D14: Case Study for KG-MHQA (5/6)

Case Study for KG-MHQA:

Verifying the Nominations: The nomination for 'The Quiet Man' aligns with the information provided, confirming that Victor Young fits the description given in the question.

Conclusion: By connecting the dots between Victor Young's place of origin, his award nominations, and the place of death for William Powell and Richard Thorpe, we conclude that Victor Young is the musician in question.

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To solve the multi-hop question, we first need to identify the place of death for both William Powell and Richard Thorpe, which is Palm Springs, California. This location is significant as it is also the place of origin for the musician we are seeking. Next, we need to determine which musician was born in Palm Springs. Victor Young, a renowned composer, fits this criterion. We then verify that Victor Young was nominated for an Academy Award for Best Original Score for 'The Quiet Man', which satisfies one of the award nomination criteria. Although the question mentions an unspecified nomination, Victor Young's career in music composition was notable, and he received multiple nominations, indicating his eligibility. Therefore, the musician who matches all the criteria in the question is Victor Young.

The final answer is: Victor Young

</output>

Figure D15: Case Study for KG-MHQA (6/6)

Cases of Two Error Types

Logical error: Logical errors in multi-hop reasoning often arise from insufficient understanding of complex problems and the incorrect planning of reasoning paths, which leads to errors in reasoning logic. For example, failing to connect all relevant facts can result in logical issues within the reasoning process. For example: *This corporation, which has an asset situated in the same city as Vivendi, is linked to a town where notable figures like Ally_Sheedy, Crispin_Glover, and Lance_Henriksen were born. What is the name of this corporation?* Answering this question requires understanding the shared asset between Vivendi and the subject corporation, and then connecting that shared asset to the birthplace of Ally_Sheedy, Crispin_Glover, and Lance_Henriksen. However, in this case, the model erroneously associates information from Document 3 and Document 4 during the decomposition of the subproblems, leading to a logical error in the first step of reasoning. Example is shown in Figure E2

Knowledge error: This type of error arises when models incorrectly select irrelevant knowledge or incorrect relevant knowledge as the final answer, leading to reasoning errors. Such mistakes typically happen when the reasoning step involves a high density of knowledge. For example, consider the question: *This show and two others share the same Netflix genre, with one featuring Meryl Streep and another starring Anil Kapoor and Mary Lynn Rajskub. What is this show?* The model correctly identifies the other two shows as Angels in America and 24, but in the final step, it selects incorrect knowledge, leading to an error in the reasoning process. Example is shown in Figure E1

Error Sample for Knowledge Error:

Question:

This show and two others share the same Netflix genre, with one featuring Meryl Streep and another starring Anil Kapoor and Mary Lynn Rajskub. What is this show?

Relevant Documents:

Document [1]: In the television show '24', Mary Lynn Rajskub and Anil Kapoor appeared as actors.

Document [2]: The shows 'Angels in America', 'Charmed', '24', and 'The Muppet Show-GB' are categorized under the genre 'Television' on Netflix.

Document [3]: In the television show 'Angels in America', Meryl Streep and Mary-Louise Parker appeared as actors.

Answer: charmed

KG-o1 Response:

<think>

Exploring the Drama Genre:

...

Conclusion:

By understanding the characteristics and scope of the 'Television' genre on Netflix, one can better appreciate the variety and depth of content available. This genre serves as a cornerstone for traditional television enthusiasts and those interested in exploring a wide range of television-based narratives.

Understanding the Netflix Genre:

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Conclusion:

Based on the analysis of the Drama genre and the known shows featuring specific actors, we can conclude that '24' is another show that shares the same Netflix genre as the first show. This conclusion is supported by the thematic elements and actor associations within the Drama genre.

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To solve the multi-hop question, we first need to identify the Netflix genre that the show in question belongs to. By examining the genre classification on Netflix, **we determine that the show is categorized under the Drama genre.** This genre is characterized by its focus on intense and emotional narratives, often involving complex character development and realistic storytelling. **Next, we identify the first show that features Meryl Streep, which is 'Angels in America'.** This show is known for its exploration of the AIDS crisis in the 1980s and is categorized under the Drama genre. **Then, we identify the second show that features Anil Kapoor and Mary Lynn Rajskub, which is '24'.** This show is a high-octane action-drama series that follows a counter-terrorism agent over the course of 24 hours and is also categorized under the Drama genre. **Finally, we confirm that the third show, which shares the same Netflix genre as the first and second shows, is indeed part of the Drama genre.** By cross-referencing the known shows featuring specific actors, **we conclude that '24' is the show that shares the same Netflix genre as the first show.** Therefore, the final answer is '24'.

The final answer is: 24

</output>

Figure E1: Sample for Knowledge Error

Error Sample for Logical Error:

Question:

This corporation, which has an asset situated in the same city as Vivendi, is linked to a town where notable figures like Ally_Sheedy, Crispin_Glover, and Lance_Henriksen were born. What is the name of this corporation?

Relevant Documents:

Document [1]:Ally Sheedy, Lance Henriksen, and Crispin Glover were born in New York City.

Document [2]:The city or town for the mailing address of Marvel Entertainment, Marvel Comics, and NBCUniversal is New York City.

Document [3]:Fairfield is the city or town for the mailing address of Fairfield University and General Electric.

Document [4]:NBCUniversal is an asset owned by Vivendi and General Electric.

...

Answer: General Electric

KG-01 Response:

<think>

Understanding Corporations and Their Assets:

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Brainstorming Possible Connections:

When considering Fairfield, it becomes apparent that no major multinational corporations are headquartered there. Thus, if a corporation has an asset in Fairfield, it would likely be one with historical ties to other cities or known for its broad geographic influence.

Exploring Media and Entertainment Companies:

...

Unpacking the Connections Between the Corporation and Fairfield:

The connection between the corporation's asset and Fairfield is crucial. This involves understanding the nature of the asset and the role of Fairfield as either part of a broader city network or as a direct asset location.

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To solve the multi-hop question, we need to identify a corporation with an asset in the same city as Fairfield and that is linked to the birthplace of notable figures Ally Sheedy, Crispin Glover, and Lance Henriksen. **The process is broken into two sub-problems.** First, we need to determine **which corporation has an asset in the same city as Fairfield.** Through research, we find NBCUniversal as a corporation that has a significant presence in New York City, a major city within proximity to Fairfield. Next, we need to establish a connection between the city where the corporation's asset is located (New York City) and **the birthplace of Ally Sheedy, Crispin Glover, and Lance Henriksen, which is Springfield.** Despite Springfield not being directly linked to NBCUniversal, the cultural and industry significance of New York City's connection to Springfield helps in understanding the broader context. Therefore, NBCUniversal meets the criteria of having an asset in a city connected to the common birthplace of these

notable figures. Thus, the name of the corporation is NBCUniversal.

The final answer is: NBCUniversal

</output>

Figure E2: Sample for Logical Error